Chapter 11

Problems 2

In this chapter we continue honing our problem-solving skills by taking on increasingly more and more challenging problems, the Three-Circle Problem Of Apollonius and several problems that deal with orthogonal circles.

Problem 14 (or the Three-Circle Problem of Apollonius). Construct all circles that are tangent to the three given (coplanar) circles, $q_1\left(Q_1, R_1\right), q_2\left(Q_2, R_2\right)$ and $q_3\left(Q_3, R_3\right)$.

Solution. Before we deep-dive into a solution of the Three-Circle Problem of Apollonius, let us first try to understand what type of solution circles are we to expect to begin with?

In general, the three-circle problem of Apollonius admits 8 distinct solution circles that can be grouped into 3 sets.

There are 2 distinct solution circles in the first set, one circle that touches all three given circles **in**ternally and another circle that touches all three given circles **ex**ternally.

The remaining 2 sets contain 3 distinct solution circles each. The circles in one such solution set touch in turn each one of the given circles **ex**ternally, while touching the remaining two given circles **in**ternally.

The circles in the other such solution set touch in turn each one of the given circles **in**ternally, while touching the remaining two given circles **ex**ternally.

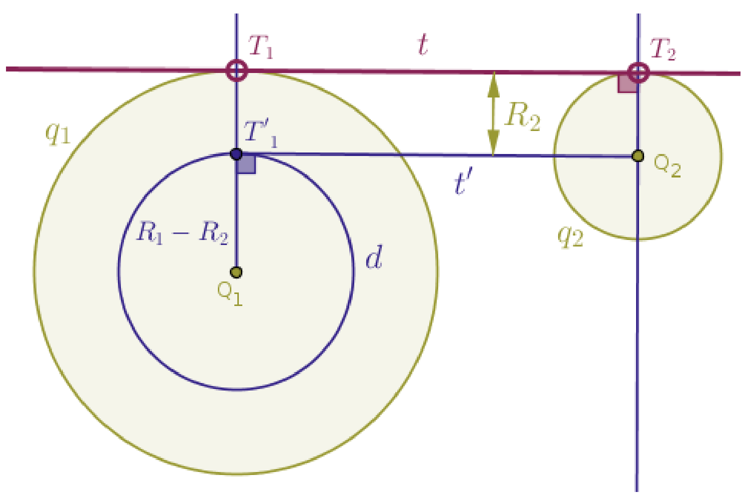
In order to recover all of these solution circles, we will employ a combination of the following two main ideas. The first idea that we will use to solve the Three-Circle Problem of Apollonius is that of reduction as we will make an attempt to reduce the current problem to a previously solved one, the Problem 12 which we solved in the previous chapter. The said reduction idea is more global or more general and is less problem-specific in nature.

The second idea that we will use to solve the Three-Circle Problem of Apollonius is similar to the idea used for constructing straight lines that play the roles of common tangent **straight lines** to two given circles. Thus, our second idea is less general and is more problem-specific.

As such, let us analyze the tactical advantage afforded by the following way of constructing straight lines that play the roles of both **ex**ternal and **in**ternal tangents to two given distinct coplanar circles of radii R_1 and R_2.

Without the loss of generality and for the sake of definitiveness we shall assume that $R_1 \geqslant R_2$, so that the geometric interpretation of the difference $(R_1 - R_2)$ remains valid and traditional.

While there are a number of distinct ways to construct a straight line that plays the role of an **ex**ternal tangent to two given distinct circles, the idea behind the method that is of interest to us is to construct an auxiliary object known as "a helper circle", d, whose center is located at the center of the larger circle, Q_1, and the length of whose radius is equal to the difference $(R_1 - R_2)$, which, as we agreed earlier, always has a meaningful geometric interpretation (Figure 11.1):

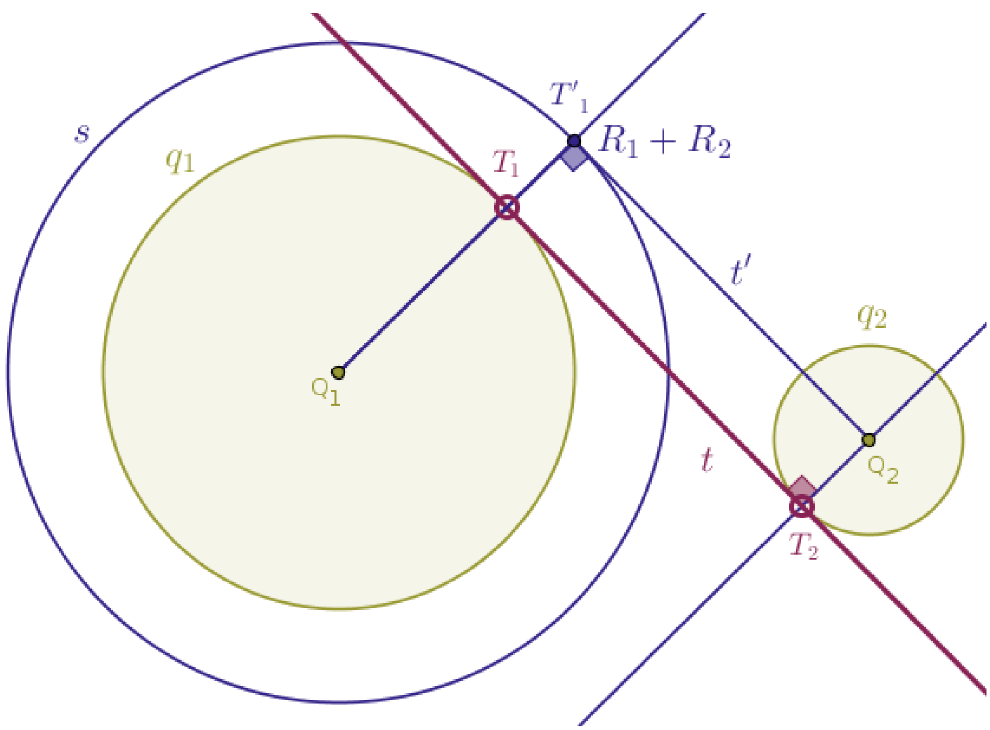


Once the circle d is drawn, we construct a straight line t' that is tangent to the said circle d and passes through the center of the smaller circle, Q_2. That particular, simpler, problem has already been solved, it is well-understood and the by-product of such a solution will be a point, T_1', where the straight line t' touches the circle d.

Technically, we may draw a straight line through the points Q_1 and T_1' until that straight line intersects the given circle q_1 at the point T_1 and then we may draw a straight line through the point Q_2 perpendicular to t' until that straight line intersects the given circle q_2 at the point T_2. Then, the straight line t through the points T_1 and T_2 will be the sought-after **ex**ternal tangent common to the given circles q_1 and q_2.

However technically correct, the above way of obtaining the sought-after tangent straight line t is not as enlightening as the idea of rigid motion known as parallel translation: the said tangent straight line t can be obtained by parallel translating the straight line t' by the distance R_2 **away from** the center of the larger circle Q_1, see the figure above. For the sake of brevity, we shall refer to the above process as the $(R_1 - R_2)$ idea.

Likewise, as the readers have no doubt figured out already, in order to construct a straight line that plays the role of a common tangent that is **in**ternal to two given circles, we will employ the equivalent $(R_1 + R_2)$ idea. That is, in order to construct a common **in**ternal tangent to two given circles, we first construct a helper circle s whose center is located at the center of the larger circle, Q_1, and whose radius is equal to the sum $(R_1 + R_2)$, which always has a meaningful geometric interpretation (Figure 11.2):



Again, once the circle s is drawn, we construct a straight line t' that is tangent to the said circle s and passes through the center of the smaller circle, Q_2.

Parallel translating the straight line t' by the distance R_2 **toward** the center of the larger circle, Q_1, recovers the sought-after straight line t that plays the role of a common tangent **in**ternal to two given circles, q_1 and q_2.

Now that we understand the essence of the tactical device used to draw the straight lines that play the roles of tangents common to two given circles, the remaining solution of the three-circle problem of Apollonius should be transparent.

Without the loss of generality we rename/renumber the given radii in such a way that the following order relations hold:

$$R_1 \geqslant R_3, \ R_2 \geqslant R_3$$

We, then, construct two auxiliary helper circles, $k_1\,(Q_1, R_1 - R_3)$ and $k_2\,(Q_2, R_2 - R_3)$, and we rephrase a previously solved problem, the Problem 12, as follows:

construct all circles that are tangent to two given circles k_1 and k_2 and that pass through the given point Q_3

But we already know how to solve the above problem:

- invert the helper circles k_1 and k_2 with respect to the circle of inversion $c\,(Q_3, r)$ centered at the point Q_3 into the circles k_1' and k_2' with positive power once
- construct two straight lines t_1 and t_2 that play the roles of common **ex**ternal tangents to the image circles k_1' and k_2'
- and invert the straight lines t_1 and t_2 in the circle c with positive power once into the circles w_1' and w_2'

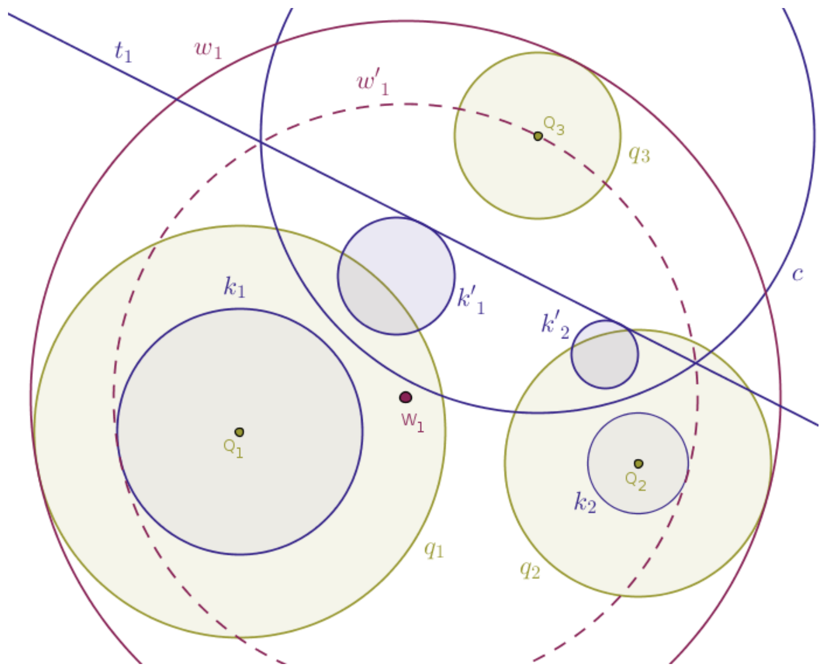
To avoid the clutter, in the figure below we show only the reflection of the above straight line t_1 in the circle c with positive power into the circle $w_1'\,\left(W_1, R_{w_1}'\right)$.

Observe that the circle w_1' thus constructed has all the important properties that we are looking for.

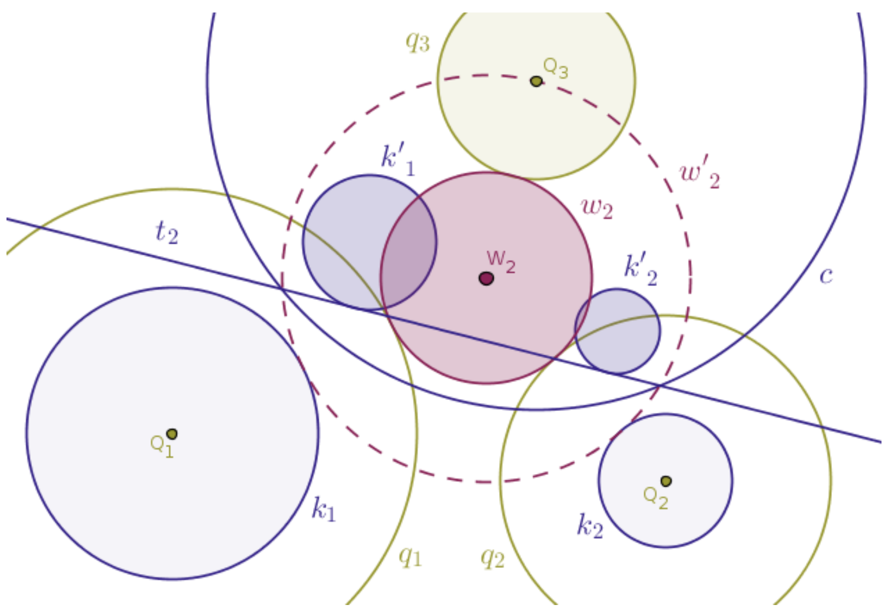
First, due to the preservation of tangency points under any inversion of the real plane with respect to a circle, since the straight line t_1 is tangent to the circles k_1' and k_2', the circle w_1' will be tangent to the inverse images of the circles k_1' and k_2'. That is, the circle w_1' will touch both helper circles k_1 and k_2 internally.

Second, the circle w_1' is concentric with the sought-after circle w_1 that touches all three given circles $q_{1,2,3}$ **in**ternally.

Therefore, all that we have to do to in order to actually recover the first solution circle w_1 is keep the center of the circle w_1' where it is, namely at the point W_1, and then we just have to **in**crease the length of the radius of the circle w_1' exactly by the length of the radius R_3 and the game is over, the circle $w_1\left(W_1, R'_{w_1}+R_3\right)$ will touch all three given circles $q_{1,2,3}$ **in**ternally (Figure 11.):



In the next drawing we show how to obtain the second solution circle, w_2, that touches all three given circles $q_{1,2,3}$ **ex**ternally (Figure 11.4):



This time, the intermediate circle $w_2' \left(W_2, R_{w_2}'\right)$ will touch the helper circles k_1 and k_2 **ex**ternally but it will also pass through the point Q_3 exactly.

Therefore, in order to obtain the sought-after solution circle w_2 that will touch all the given circles $q_{1,2,3}$ **ex**ternally, all that we have to do is keep the center of that circle at the point W_2 and then we simply have to **de**crease the length of the radius of the circle w_2' by exactly the length of the radius of the third circle R_3 and the circle $w_2 \left(W_2, R_{w_2}' - R_3\right)$ will touch all three given circles $q_{1,2,3}$ **ex**ternally.

Thus, we may say intuitively that in order to recover the above two solution circles w_1 and w_2 we have used the $(R_1 - R_2)$ idea.

It stands to reason, that in order to obtain the remaining six solution circles $w_{3\text{–}8}$ we, intuitively speaking, will employ the $(R_1 + R_2)$ idea.

Since all of the remaining six solution circles $w_{3\text{–}8}$ are obtained in exactly the same way, we will show how to obtain two such solution circles, $w_{3,4}$, and our readers are encouraged to construct the remaining solution circles by themselves.

As such, we will obtain:

- the solution circle w_3 that touches the given circle q_1 **ex**ternally, while touching the remaining two given circles $q_{2,3}$ **in**ternally and
- the solution circle w_4 that, conversely, touches the given circle q_1 **in**ternally, while touching the remaining two given circles $q_{2,3}$ **ex**ternally

To that end, construct the helper circles $k_2\,(Q_2, R_2 + R_1)$ and $k_3\,(Q_3, R_3 + R_1)$ and reflect both of these circles in a circle $c\,(Q_1, r)$ with positive power, once, into the circles k_2' and k_3'.

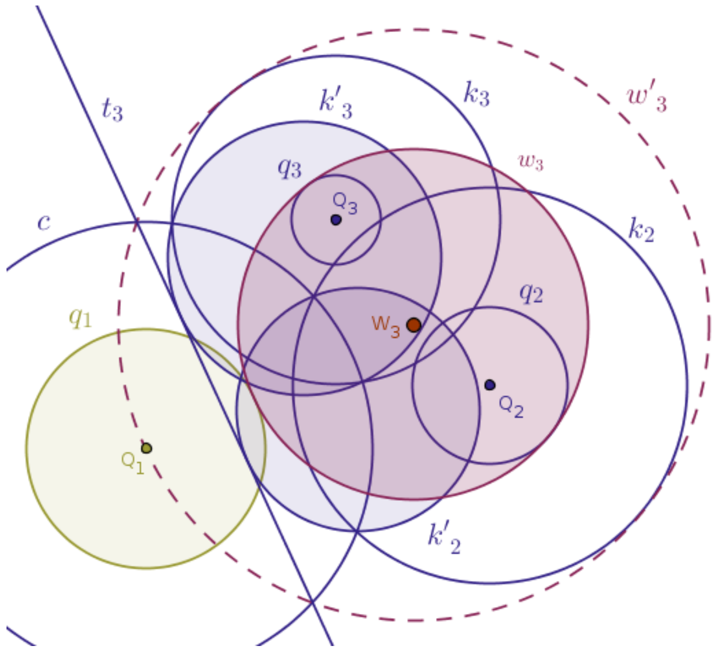
Draw two straight lines, t_3 and t_4, that play the roles of common **ex**ternal tangents to the circles k_2 and k_3.

Invert the straight line t_3 in the same circle $c\,(Q_1, r)$ with positive power, once, into an intermediate circle, $w_3' \left(W_3, R_{w_3}'\right)$, that passes through the point Q_1 and touches both of the helper circles k_2 and k_3 **in**ternally.

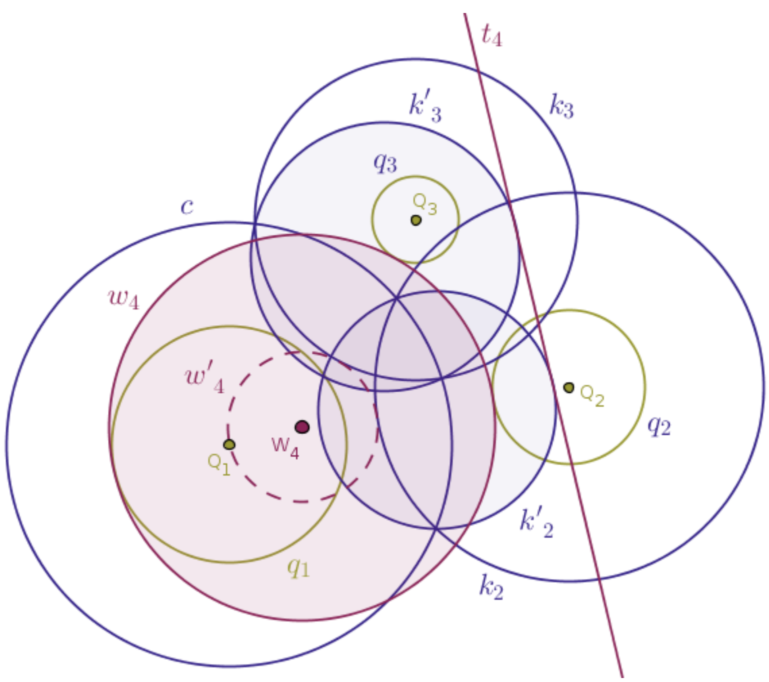
By construction, that intermediate circle will be concentric with the sought-after, solution circle w_3 that will touch the circle q_1 externally, while touching the circles $q_{2,3}$ internally.

Keep the center of the intermediate circle w_3' where its, at the point W_3 that is, and simply **de**crease the radius of the said circle w_3' exactly by the length of the radius R_1.

Then, the sought-after solution circle $w_3\left(W_3, R'_{w_3} - R_1\right)$ will, as promised, touch the given circle q_1 externally, while touching the remaining two given circles, $q_{2,3}$, internally (Figure 11.5):



Likewise, invert the straight line t_4 in the same circle $c\left(Q_1, r\right)$ with positive power, once, into the circle $w'_4\left(W_4, R\prime_{w_4}\right)$ that passes through the point Q_1 and touches both of the helper circles k_2 and k_3 externally (Figure 11.6):



Keep the center of the circle w_4' where its, at the point W_4 that is, and simply increase the radius of the said circle w_4' exactly by the length of the radius R_1. The sought-after solution circle $w_4\left(W_4, R'_{w_4} + R_1\right)$ thus constructed will touch the given circle q_1 internally, while touching the remaining two given circles, $q_{2,3}$, externally.

We, thus, obtain $2 + 2 = 4$ distinct solution circles so far.

By repeating the above steps for the circles q_2 and q_3, we shall recover $2 + 2 = 4$ more distinct solution circles $w_{5\text{–}8}$, bringing the total count of the distinct solution circles of the three-circle problem of Apollonius to $4 + 4 = 8$, as claimed. □

Orthogonal Circles

Problem 15. Construct a circle that is orthogonal to a given circle, $q(Q, R)$, and that passes through two given fixed distinct coplanar points, P_1 and P_2.

Solution. The two given points, P_1 and P_2, can be located in the plane that houses the given circle, q, in any number of ways: these two points can be located both **in**side or both **out**side of the given circle, they can be both located exactly on the given circle, they can be collinear with the center, Q, of the given circle and so on.

We, naturally, want to consider the mutual juxtaposition of the given points with respect to the given circle when interesting or non-trivial solutions exist.

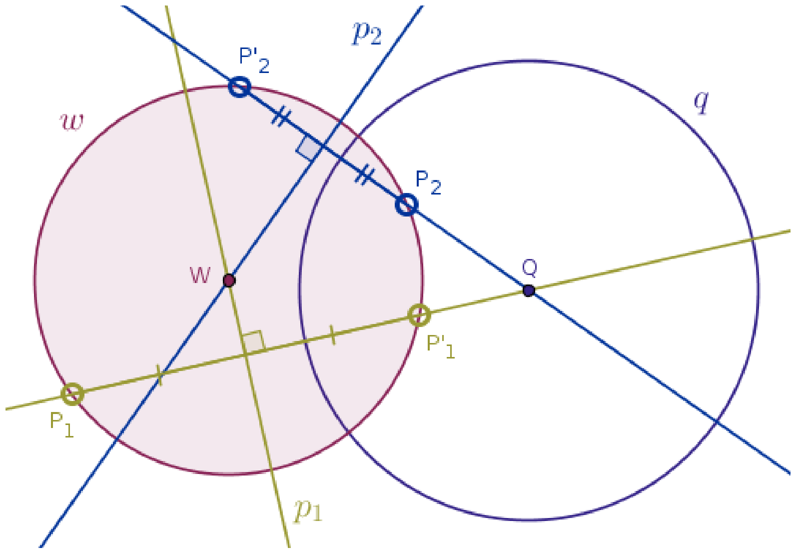
When, for example, one given point, say, the point P_1, is located **out**side of the given circle, while the remaining point, P_2, is located **in**side of the given circle.

We remember that due to the Theorem 39, it is possible to construct an infinitude of circles that are both orthogonal to the given circle, q, and pass through the given point, P_1.

To that end, we invert the point P_1 in the given circle, q, with positive power, once, into its image point P_1' and recall that the locus of the centers of all the circles that are orthogonal to the given circle, q, and pass through the given point, P_1, is the perpendicular bisector, p_1, of the line segment P_1P_1'.

Likewise, if we invert the given point P_2 with respect to the given circle, q, with positive power, once, into its image point, P_2', then the infinitude of the circles that are orthogonal to the given circle, q, and pass through the given point P_2 will have their centers located on the perpendicular bisector, p_2, of the line segment P_2P_2'.

As such, using the method of intersecting loci, we observe that since in the Euclidean plane the two distinct non parallel straight lines p_1 and p_2 intersect at a single point, name that point W, it follows that the circle w that has its center located at the intersection point W and that passes through all four points P_1, P_1', P_2 and P_2' will be the unique solution sought-after (Figure 11.7):



Thus, the non trivial solution of this particular mutual juxtaposition of the input objects is:

- invert the given points, P_1 and P_2, with respect to the given circle, q, with positive power, once, into their image points, P_1' and P_2' correspondingly
- bisect the line segments P_1P_1' and P_2P_2' with their perpendicular bisectors, p_1 and p_2 correspondingly
- let W be the point where the straight lines p_1 and p_2 intersect
- draw a circle, w, whose center is located at the point W and the length of whose radius is equal to $|WP_1| = |WP_1'| = |WP_2| = |WP_2'|$

The circle w thus constructed is the unique solution circle sought-after, a circle that passes through the given points $P_{1,2}$ and that is orthogonal to the given circle q. Proof: exercise. $\square$

Problem 16. Construct a circle that is orthogonal to two given non-concentric circles, q_1 (Q_1, R_1) and q_2 (Q_2, R_2), and that passes through a given point, P.

Solution. Here we will separately consider the three cases when the given circles q_1 and q_2:

1. are not in contact
2. touch, at a single point, T
3. intersect, at two distinct points, F_1 and F_2

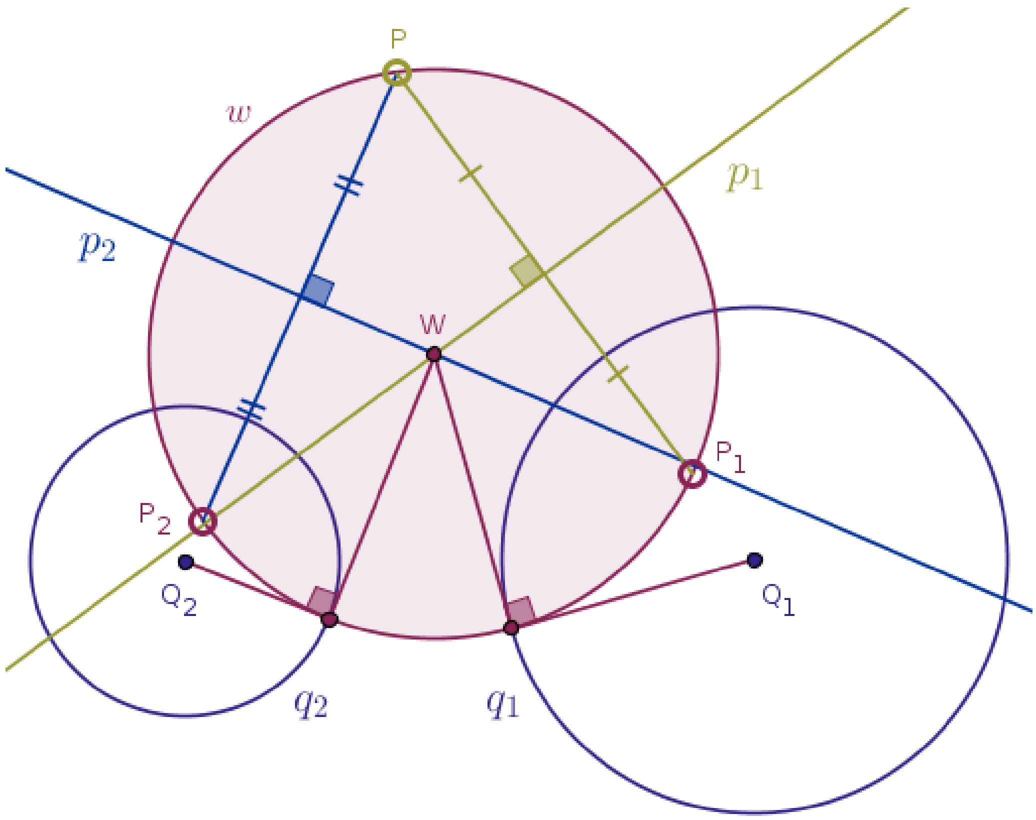
Case 1. The given circles, q_1 and q_2, are not in contact.

It is interesting that the solution of this case mimics the solution of the previous problem, the Problem 15, with one minor adjustment: instead of reflecting two distinct points in the same circle, here we invert the same point in two distinct circles.

Namely, invert the given point, P, in the given circle q_1 with positive power, once, into its image point, P_1. Invert the same given point in the given circle q_2 with positive power, once, into its image point, P_2.

Bisect the line segment PP_1 with a straight line p_1. Bisect the line segment PP_2 with a straight line p_2. Let W be the point where the straight lines p_1 and p_2 intersect.

Then, the circle w whose center is located at the point W and the length of whose radius is equal to $|WP| = |WP_1| = |WP_2|$ is the unique solution circle sought-after (Figure 11.8):



A proof that this construction is correct is, pretty much, baked right into the way in which we solved this case.

Case 2. The given circles, q_1 and q_2, touch, at a single point, T.

We shall consider the case when the given circles touch each other **ex**ternally.

If the given point, P, is indistinguishable from the point of tangency, T, that is if $P \equiv T$, then this problem has uncountably infinitely many solutions because any circle whose center is located on the straight line that is tangent to both given circles and passes through the said point of tangency, T, will be orthogonal to both input circles, q_1 and q_2.

Thus, assume that it is the case that $P \not\equiv T$.

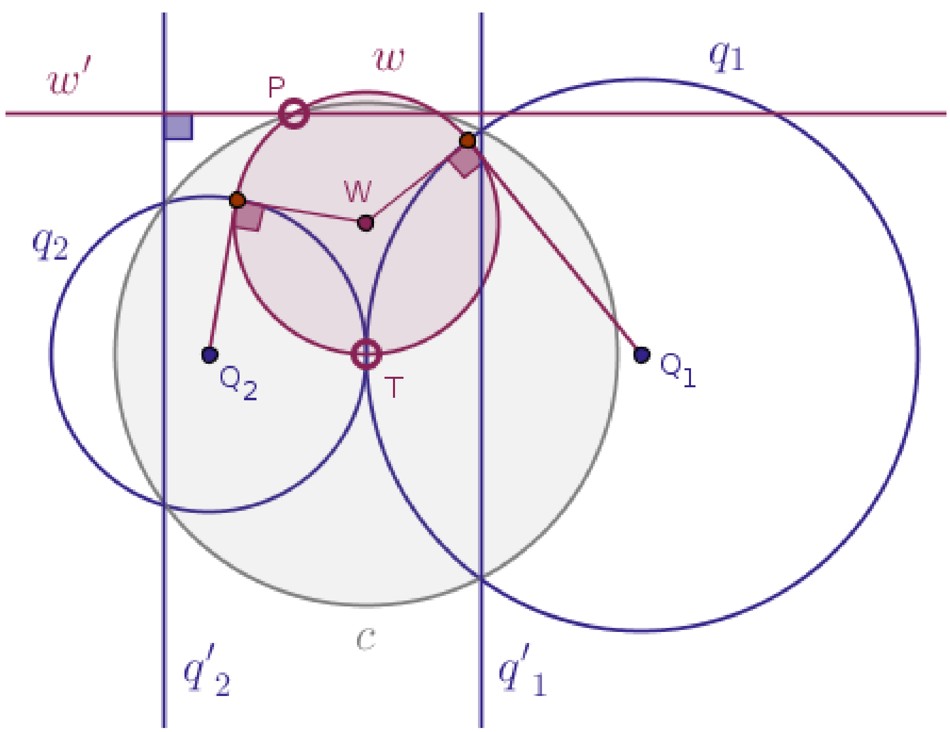
Then it makes sense to choose the circle $c(T, |TP|)$ to be the circle of inversion because we remember that due to the Theorem 15, under the inversion of the parent plane with respect to the circle c with positive power our input circles q_1 and q_2 will be reflected into straight lines, q_1' and q_2' correspondingly.

But then our inverted solution must be either a circle or a straight line that is simultaneously both orthogonal to two distinct parallel straight lines, q_1' and q_2', and passes through a given point, P.

Such a solution cannot possibly be a circle (explain why on your own).

Therefore, the only choice for the potential inverted solution is a straight line.

In the Euclidean space such a straight line, name it w', that is both orthogonal to two distinct parallel straight lines, q_1' and q_2', and passes through a given point, P, is unique (Figure 11.9):



Hence, inverted in the circle $c(T, |TP|)$ with positive power, the straight line w' will be transformed into the unique sought-after solution circle, $w(W, |WT| = |WP|)$.

Case 3. The given circles, q_1 and q_2, intersect, at two distinct points, F_1 and F_2.

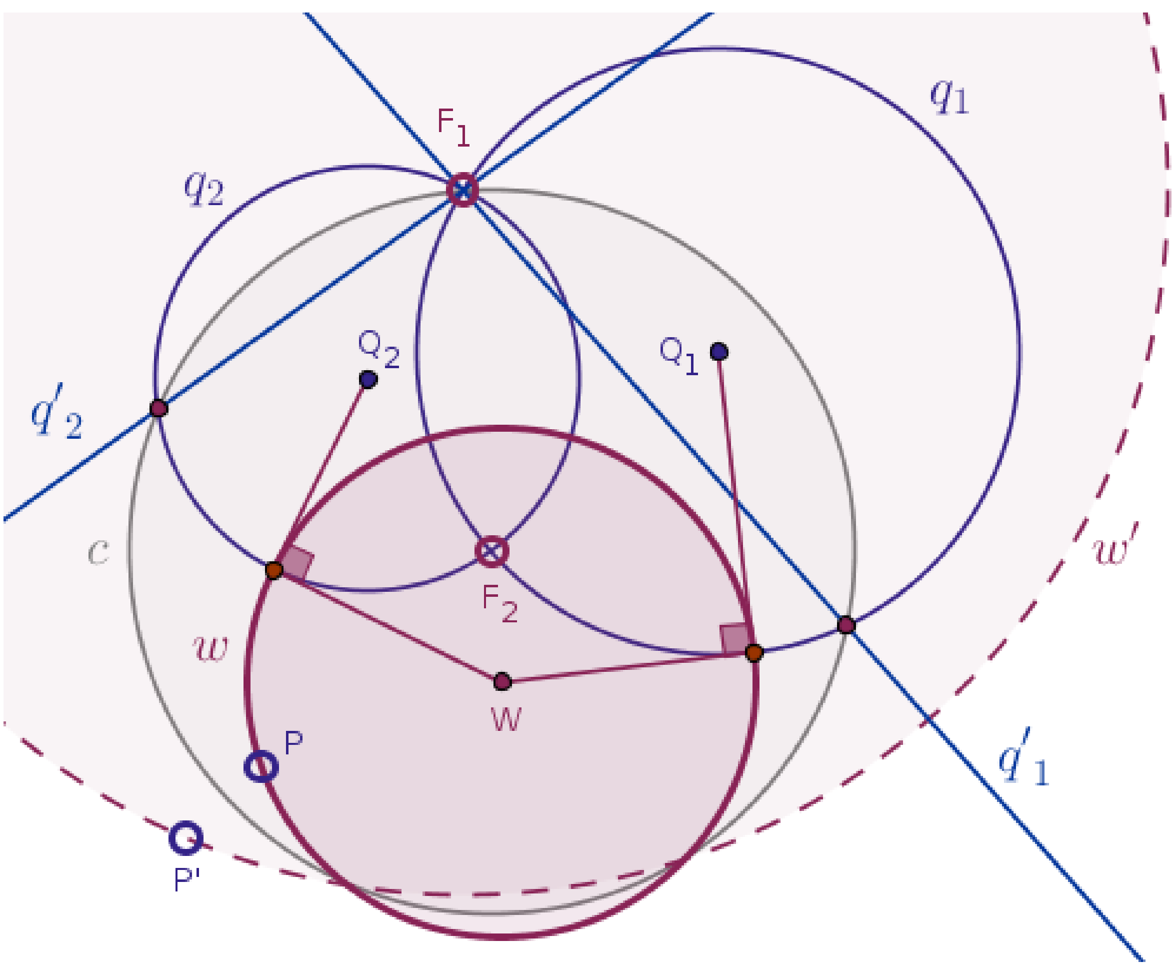
If it is the case that $P \equiv F_1$ or $P \equiv F_2$ then the sought-after solution circle exists only if the radius of one of the given circles is, trivially, zero.

Assume that it is the case that $P \not\equiv F_1$ and $P \not\equiv F_2$.

In that case it makes sense to choose either the circle $c(F_1, |F_1F_2|)$ or the circle $c(F_2, |F_2F_1|)$ as the circle of inversion. Say, we make $c(F_2, |F_2F_1|)$ our circle of inversion.

Then, under the inversion of the parent plane with respect to the circle $c(F_2, |F_2F_1|)$ with positive power, as we remember, the given circles q_1 and q_2 will be taken into the straight lines q_1' and q_2' correspondingly.

Moreover, the two distinct straight lines q_1' and q_2' will intersect at the given point F_1 and, under the same inversion, the given point, P, will be taken into its image point, P' (Figure 11.10):



But then the inverted solution in this particular case must be either a straight line or a circle that must simultaneously pass through the point P' and be orthogonal to two distinct non-parallel straight lines.

In the Euclidean space such a solution object cannot possibly be a straight line (explain why on your own).

Therefore, the only choice for the potential inverted solution is a circle.
In the Euclidean space, a Euclidean plane to be exact, such a circle, name it w', that has the given point F_1 as its center and the distance $|F_1P'|$ as the length of its radius, is unique.

Therefore, the solution object of the original problem, which we obtain by inverting the circle $w'(F_1, |F_1P|)$ in the circle $c(F_1, |F_1F_2|)$ with positive power, will also be unique.

By hypothesis, the given points P and F_2 are distinct. We also know that under no inversion of the parent plane with respect to a circle an original point distinct from the center of inversion is never taken into the said center of inversion.

Hence, all three points F_2, P and P' will be distinct. Therefore, due to the family of the previously proven theorems, numbered 22, 25, 26, 29 and 30, the solution object of the original problem will always be **a circle** and we name that solution the circle $w(W, |WP|)$. $\square$

Problem 17. Find a circle of inversion with respect to which any two distinct circles, $q_1\ (Q_1, R_1)$ and $q_2\ (Q_2, R_2)$, that are not in contact can be inverted into two concentric circles.

Solution. Luckily, in this problem there are only two possible cases of the mutual juxtaposition of the given circles with respect to each other when that problem has a meaningful solution:

1. both given circles are located one **out**side of the other
2. one of the given circles is located **in**side of the other

Case 1. Both given circles are located one **out**side of the other.

Without the loss of generality we will assume that it is the case that $R_1 \geqslant R_2$, since otherwise we just rename the circles in such a way that the last order relation holds.

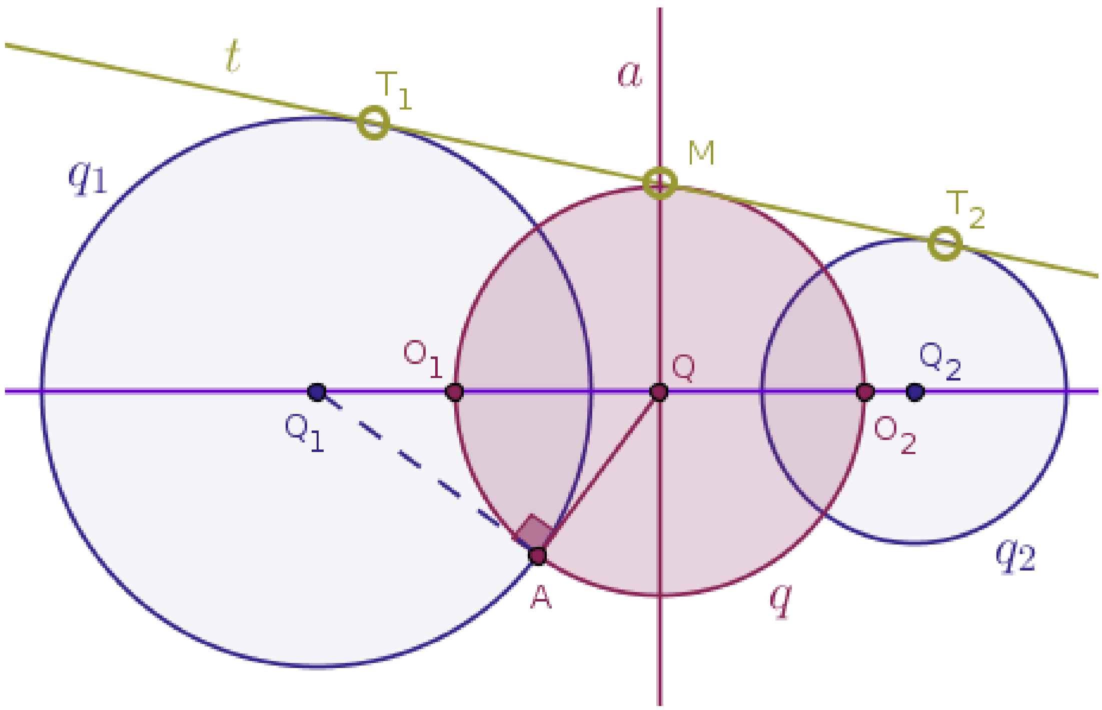
From the Case 3 of the Radical Axis section of the Chapter 9 of this book on page 134, we remember that the radical axis, a, of the input circles q_1 and q_2 is the locus of the centers of all the circles that are orthogonal to both of these circles, q_1 and q_2.

Solving this problem in reverse order, we, temporarily, assume that the radical axis of the given circles is already constructed.

Draw a straight line through the centers of the given circles, Q_1 and Q_2, until that straight line intersects the said radical axis, a, at the point Q.

Through the point Q draw a straight line that is tangent to the circle, say, q_1 at the point, say, A. That task can be carried out with the Euclidean tools alone.

Next, draw a circle $q(Q, |QA|)$ centered at the point Q with radius whose length is equal to the distance between the points Q and A. As was shown earlier, the circle q will be orthogonal to both circles q_1 and q_2 (Figure 11.11):



Let the straight line of given centers Q_1Q_2 intersect the circle q at:

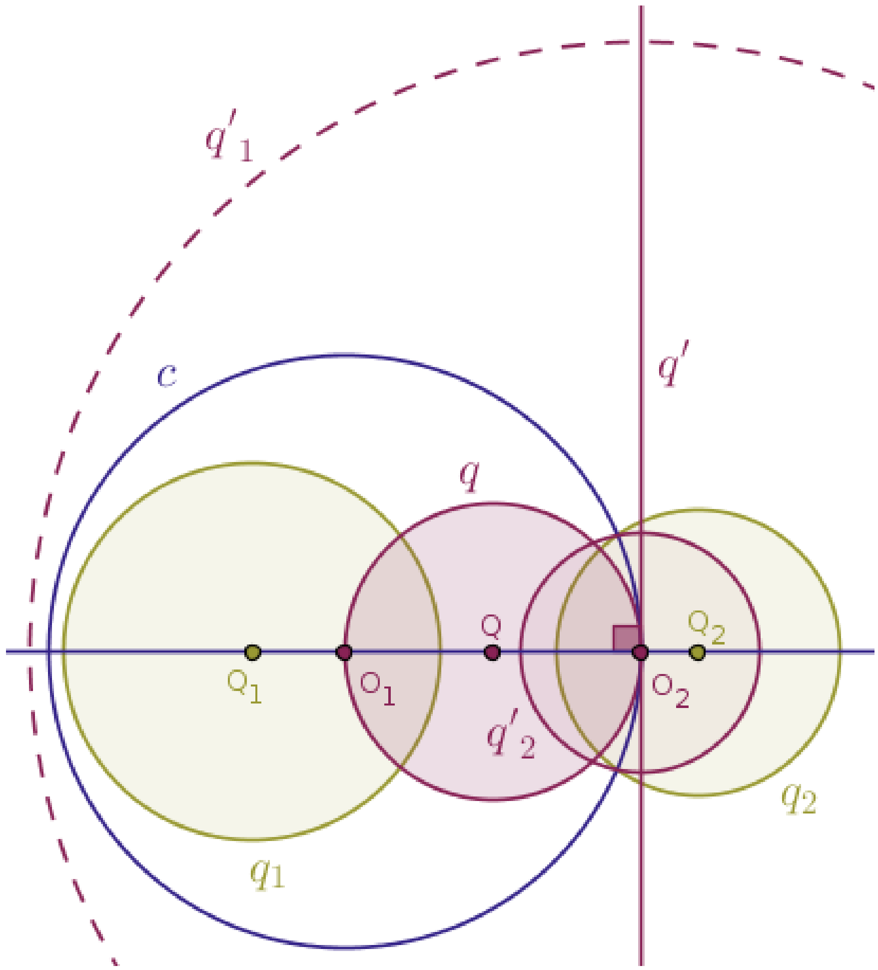
- the point O_1 on the side of the center Q_1 of the circle q_1 and
- the point O_2 on the side of the center Q_2 of the circle q_2

Then, the solution circle sought-after will be either the circle $c(O_1, |O_1O_2|)$ centered at the point O_1 or the circle $c(O_2, |O_2O_1|)$ centered at the point O_2.

Indeed, let us see where choosing the circle $c(O_1, |O_1O_2|)$ as the circle of inversion will take us.

Under the inversion of the parent plane with respect to the circle $c(O_1, |O_1O_2|)$ with positive power, due to the Theorem 9, the straight line of centers Q_1Q_2 will be taken into itself.

Under the same inversion, due to the Theorem 17, the circle q that passes through the center of inversion, O_1, and touches the circle of inversion internally at the point O_2 will be taken into a straight line, q', that is tangent to the circle c at the point O_2 (Figure 11.12):



Which is to say that the straight line q' will be orthogonal to the straight line Q_1Q_2, since the original objects, the circle q and the straight line Q_1Q_2, are orthogonal.

Next, we remember that under any inversion of the real plane with respect to a circle the angles between the curves are preserved due to the collection of the theorems 32–35. Therefore, if the circles q_1 and q_2 are both orthogonal to the straight line Q_1Q_2 then both images of these circles under the inversion of the parent plane with respect to the circle $c(O_1, |O_1O_2|)$ with positive power will also be orthogonal to the straight line Q_1Q_2.

Due to the previously proven theorems, it follows that the said images of the circles q_1 and q_2 will be circles also . Let us name such circles as q_1' and q_2' correspondingly.

By the theorems numbered 22 through 30, the centers of the said circles q_1' and q_2' must be located somewhere on the straight line Q_1Q_2.

Moreover, since the circle q is orthogonal to the circles q_1 and q_2, under the inversion of the parent plane with respect to the circle $c(O_1, |O_1O_2|)$ with positive power its image, the straight line q', must also be orthogonal to the circles q_1' and q_2'. For the latter orthogonality requirement to hold, the centers of the circles q_1' and q_2' must be located somewhere on the straight line q'.

The only location that satisfies both of the above requirements for the centers of the circles q_1' and q_2' is the point where the straight lines Q_1Q_2 and q intersect, the point O_2, which, in turn, means that the circles q_1' and q_2' are concentric, as demanded.

Collecting the above reasoning into a set of solution steps, in order to find a circle of inversion with respect to which any two distinct circles, $q_1\ (Q_1, R_1)$ and $q_2\ (Q_2, R_2)$, that are not in contact and that are located one outside of the other are reflected into two concentric circles:

- draw the straight line of centers, Q_1Q_2, of the given circles
- draw one external tangent straight line that touches the circle q_1 at the point T_1 and the circle q_2 - at the point T_2
- bisect the line segment T_1T_2 in order to locate its middle point M such that $|MT_1| = |MT_2|$
- draw a straight line, a, perpendicular to the straight line Q_1Q_2 through the point M in order to locate the point of their intersection, the point Q
- draw a straight line that passes through the point Q and that is tangent to the circle q_1 at the point A
- draw a circle $q(Q, |QA|)$ centered at the point Q with the radius whose length is equal to the distance between the points Q and A
- let O_1 and O_2 be the points where the circle q and the straight line Q_1Q_2 intersect
- the circle $c(O_1, |O_1O_2|)$ is a circle of inversion with positive power sought-after

The investigation of the case when one of the given circles, q_1 or q_2, is located inside of the other is left to the reader as an exercise. $\square$